

PROJECT SYNOPSIS

Thermal hotspot detection in solar panel using AI and infrared imaging

1. Introduction

Solar photovoltaic installations lose a meaningful share of their output every year to problems that are invisible to the naked eye. A cracked cell, a failing bypass diode, a loose junction, or a patch of shading can quietly push a section of a panel into overheating, and by the time the drop in generation is noticed on a meter, the underlying fault has often been active for weeks. Thermal imaging has long been used to catch this kind of damage, but in most solar farms it still depends on a technician walking the array with a handheld FLIR camera or flying a drone over it on a fixed schedule. Both approaches are useful, but neither one watches the panels continuously, and both rely on a person to look at the thermal image and decide whether what they are seeing is actually a fault.

This project builds a compact, low-cost monitoring unit that removes that dependency on manual inspection. A Raspberry Pi 4 is paired with a low-resolution thermal camera and a standard visual camera, and the two image streams are combined and fed into a lightweight object detection model that has been trained to recognise the thermal signature of a hotspot. The idea is not to replace professional-grade thermography equipment, but to give a single panel, or a small residential or campus-scale array, a way to flag problems on its own, in real time, without needing a technician to be physically present.

2. Objectives

- Build a portable hardware unit that can capture synchronized thermal and visual images of a solar panel.
- Process the low-resolution thermal data (32x24 pixels) into a usable higher-resolution thermal map through interpolation.
- Align the thermal and visual frames so that a detected hotspot can be located precisely on the physical panel.
- Train and deploy a YOLOv8-nano model to automatically detect and localize hotspot regions.
- Correlate detected hotspots with electrical readings (voltage/current) to link thermal anomalies with actual performance loss.
- Display results on-device via a touchscreen for immediate, on-site feedback.

3. Scope of the Project

The system is designed as a proof-of-concept prototype rather than a production monitoring solution for utility-scale solar farms. It targets small to medium installations — rooftop arrays, campus solar projects, or individual panel testing during quality checks — where a low-cost, self-contained unit can be mounted or handheld to periodically scan panels. The project covers the hardware assembly, the image processing pipeline, and the detection model, but does not extend into networked fleet monitoring or cloud-based dashboards, which are noted as possible future extensions.

4. Proposed Methodology

The system follows a four-stage pipeline. First, data acquisition and interpolation: the MLX90640 captures a raw 32x24 thermal frame, which is upsampled through interpolation to produce a smoother, more usable thermal image, while the Pi Camera simultaneously captures a visual frame of the same panel. Second, preprocessing and thermal-visual registration: since the two cameras sit at slightly different positions and angles, their frames are geometrically aligned so a coordinate on the thermal map corresponds correctly to the same physical point in the visual image, and the temperature values are normalized for consistent input to the detection stage. Third, computer vision hotspot localization: the aligned thermal data is scanned for regions of abnormal temperature relative to the surrounding cells. Fourth, detection with YOLOv8-nano: a lightweight, Pi-friendly version of the YOLO object detection architecture is used to classify and draw bounding boxes around confirmed hotspot regions, chosen specifically because it is small enough to run inference on a Raspberry Pi 4 without external acceleration.

Alongside the imaging pipeline, the INA219 sensor logs the panel's voltage and current so that a detected hotspot can be cross-checked against an actual dip in electrical output, and the DS18B20 provides an ambient/contact temperature reference to help distinguish a genuine fault from ordinary heating caused by sunlight and weather conditions.

5. Hardware Components

Component	Specification
Thermal Sensor	MLX90640 Thermal Camera Module (32 x 24 IR array)
Processing Unit	Raspberry Pi 4 Model B (4GB / 8GB RAM)
Visual Camera	Raspberry Pi Camera Module 3 (12 MP)

Display	3.5-inch / 5-inch HDMI Touchscreen
Storage	32GB microSD Card, Class 10
Power Supply	USB-C Power Bank
Auxiliary Sensors	INA219 Voltage/Current Sensor, DS18B20 Temperature Sensor

6. Expected Outcome

The finished prototype should be able to scan a panel, flag any region running noticeably hotter than its surroundings, and display the location of that hotspot on-screen alongside the corresponding visual image, within a few seconds of capture. If the electrical correlation stage works as intended, the system should also be able to indicate whether a detected hotspot is coinciding with a measurable drop in the panel's output, which would make the tool useful not just for spotting damage but for estimating its severity.

7. Conclusion

This project is an attempt to make thermal fault detection in solar panels less dependent on manual inspection, using hardware that is inexpensive enough to be replicated widely. By combining a thermal sensor, a standard camera, and a small object detection model on a single Raspberry Pi, it aims to demonstrate that early-stage automated fault detection is achievable without expensive industrial thermography equipment.